

Lecture 08 - Local Illumination

This lecture explains how visible surface points receive color from local light, material, normal, and view information. It is about shading a point, not solving all global light transport.

Source chunk	course_text_parts/03_lectures/08_local-illumination.txt
Extracted slide pages	62
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Per-slide commentary

Each extracted slide page gets its own note. Use this section as the spoken commentary that the terse slide deck is missing.

Page 1	Untitled slide
Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Commentary	This slide is about Untitled slide. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Untitled slide' into a causal sentence instead of repeating the slide title?
Page 2	Local Illumination
Source cue	- Illumination is the process of simulating light interactions in a virtual scene / - Requires light and material properties / - Achieved through illumination model / - Essential for scene perception / without illumination with illumination
Commentary	This slide is about Local Illumination. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Illumination is the process of simulating light interactions in a virtual scene / - Requires light and material properties / - Achieved through illumination model / - Essential for scene perception / without illumination with illumination
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Local Illumination'?

Page 3	8.1 Physics of Light
Source cue	8.2 Light Sources / 8.3 Material Models / 8.4 Phong Illumination Model / 8.5 Shading
Commentary	This slide is about 8.1 Physics of Light. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: 8.2 Light Sources / 8.3 Material Models / 8.4 Phong Illumination Model / 8.5 Shading
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '8.1 Physics of Light'?
Page 4	8.1 Physics of Light
Source cue	Understanding light in the real world
Commentary	This slide is about 8.1 Physics of Light. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: Understanding light in the real world
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '8.1 Physics of Light'?
Page 5	Electromagnetic Radiation
Source cue	- Electromagnetic radiation is defined by a spectrum of wavelengths / Wave length λ / - Visible light is subset of electromagnetic spectrum between 400 and 700nm, / which humans perceive as colors
Commentary	This slide is about Electromagnetic Radiation. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Electromagnetic radiation is defined by a spectrum of wavelengths / Wave length λ / - Visible light is subset of electromagnetic spectrum between 400 and 700nm, / which humans perceive as colors
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Electromagnetic Radiation'?
Page 6	Visible Light
Source cue	1000 nm / Infrared / Radio / 700 nm / TV Red
Commentary	This slide is about Visible Light. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: 1000 nm / Infrared / Radio / 700 nm / TV Red
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.

Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Visible Light'?
Page 7	Visible Light
Source cue	1000 nm / Infrared / Radio / 700 nm / TV Red
Commentary	This slide is about Visible Light. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: 1000 nm / Infrared / Radio / 700 nm / TV Red
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Visible Light'?
Page 8	Light Spectra
Source cue	- Different light sources have different light spectra / - Intensity differs based on wavelength in visible spectrum / [http://www.micro.magnet.fsu.edu/optics/lightandcolor/sources.html]
Commentary	This slide is about Light Spectra. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Different light sources have different light spectra / - Intensity differs based on wavelength in visible spectrum / [http://www.micro.magnet.fsu.edu/optics/lightandcolor/sources.html]
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Spectra'?
Page 9	Black-Body Radiation
Source cue	- A continuous spectrum emitted by a black body at a certain temperature / Steel / > 1570 K / 820 K / Stars
Commentary	This slide is about Black-Body Radiation. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - A continuous spectrum emitted by a black body at a certain temperature / Steel / > 1570 K / 820 K / Stars
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Black-Body Radiation' into a causal sentence instead of repeating the slide title?
Page 10	Black-Body Radiation
Source cue	Stars / https://www.tec-science.com/thermodynamics/temperature/black-body-radiation/ 23/06/2026 10

Commentary	This slide is about Black-Body Radiation. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: Stars / https://www.tec-science.com/thermodynamics/temperature/black-body-radiation/ 23/06/2026 10
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Black-Body Radiation' into a causal sentence instead of repeating the slide title?

Page 11	Spectral Density Function (SDF)
Source cue	- Light spectra are represented as spectral density functions / - Spectral density functions provide the light intensity (=energy) per wavelength / 400 Wavelength λ (nm) 700 / ygrenE / 400 Wavelength λ (nm) 700
Commentary	This slide is about Spectral Density Function (SDF). Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Light spectra are represented as spectral density functions / - Spectral density functions provide the light intensity (=energy) per wavelength / 400 Wavelength λ (nm) 700 / ygrenE / 400 Wavelength λ (nm) 700
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Spectral Density Function (SDF)'?

Page 12	Light Perception
Source cue	- Humans perceive light spectra / as differently colored lights / - Often described using three / main properties (HSL/HSV model) / - Hue - primary color tone
Commentary	This slide is about Light Perception. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Humans perceive light spectra / as differently colored lights / - Often described using three / main properties (HSL/HSV model) / - Hue - primary color tone
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Perception'?

Page 13	Interpreting SDFs
Source cue	- Shape of an SDF depicts how humans perceive represented light / - Hue: dominant wavelength / - Saturation: % of energy in dom. wavelength / - Luminance: total energy (SDF integral) / 400 Wavelength λ (nm) 700

Commentary	This slide is about Interpreting SDFs. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Shape of an SDF depicts how humans perceive represented light / - Hue: dominant wavelength / - Saturation: % of energy in dom. wavelength / - Luminance: total energy (SDF integral) / 400 Wavelength λ (nm) 700
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Interpreting SDFs'?

Page 14	Representing SDFs
Source cue	- In spectral rendering SDFs are discretized as arrays / - Cell number represents differentiable wavelength ranges / - Cell values represent energy per wavelength range / - Much more frequently SDFs are represented as three-channel color / - HSL values are derived from SDF
Commentary	This slide is about Representing SDFs. Read it as an occlusion decision. Decide whether the method reasons about objects, image regions, rays, or per-fragment depth comparisons. The visible cue is: - In spectral rendering SDFs are discretized as arrays / - Cell number represents differentiable wavelength ranges / - Cell values represent energy per wavelength range / - Much more frequently SDFs are represented as three-channel color / - HSL values are derived from SDF
Why it matters	Visibility decides which generated candidates are actually seen from the current viewpoint.
Check yourself	Can you decide whether 'Representing SDFs' works per object, per image region, per ray, or per fragment?

Page 15	8.2 Light Sources
Source cue	How to represent light sources in computer graphics
Commentary	This slide is about 8.2 Light Sources. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: How to represent light sources in computer graphics
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '8.2 Light Sources'?

Page 16	Light Sources
Source cue	- All light sources have a RGB color property representing their SDF / - Light sources are also defined through their location, orientation and extent / within a virtual scene / - Different light source types omit different spatial properties / - Light sources have no associated geometry

Commentary	This slide is about Light Sources. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - All light sources have a RGB color property representing their SDF / - Light sources are also defined through their location, orientation and extent / within a virtual scene / - Different light source types omit different spatial properties / - Light sources have no associated geometry
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Sources'?

Page 17	Point Light Sources
Source cue	- Point light sources $\mathbf{L} = \mathbf{L}$, $\mathbf{L}$ have a geometric center $\mathbf{L} = (x, y, z)$ / and radiate light with color $\mathbf{L} = (\mathbf{L}, \mathbf{L}, \mathbf{L})$ evenly in all directions
Commentary	This slide is about Point Light Sources. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Point light sources $\mathbf{L} = \mathbf{L}$, $\mathbf{L}$ have a geometric center $\mathbf{L} = (x, y, z)$ / and radiate light with color $\mathbf{L} = (\mathbf{L}, \mathbf{L}, \mathbf{L})$ evenly in all directions
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Point Light Sources'?

Page 18	Cone Light Sources
Source cue	- Special point light source with geometric center / - Radiation of light restricted to a cone-shaped area with its tip in the center / - Specified by cut-off angle and exponents for the decrease / of brightness with distance to the center / (spot light exponent)
Commentary	This slide is about Cone Light Sources. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Special point light source with geometric center / - Radiation of light restricted to a cone-shaped area with its tip in the center / - Specified by cut-off angle and exponents for the decrease / of brightness with distance to the center / (spot light exponent)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Cone Light Sources'?

Page 19	Light Attenuation
Source cue	- Light attenuation is inversely proportional to the square / of the distance $\mathbf{L}$ between the light source and the object / - Attenuation factor / $\mathbf{L} = \min , 1 / \mathbf{L}^2 + \mathbf{L} \cdot \mathbf{L} + \mathbf{L} \cdot \mathbf{L}^2$

Commentary	This slide is about Light Attenuation. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Light attenuation is inversely proportional to the square / of the distance ■ between the light source and the object / - Attenuation factor / ■ = min , 1 / ■■■ ■ + ■ · ■ + ■ · ■ ²
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Attenuation'?
Page 20	Area Light Sources
Source cue	- Light source with geometric center and geometric extension / - Mostly of planar shape / - Transmission of light over the entire surface analogous to a point light source
Commentary	This slide is about Area Light Sources. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Light source with geometric center and geometric extension / - Mostly of planar shape / - Transmission of light over the entire surface analogous to a point light source
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Area Light Sources'?
Page 21	Directed Light
Source cue	- Infinity-far away imaginary light source / - Light rays run parallel and evenly throughout the scene, / direction is specified as a vector / - Light intensity does not diminish / with increasing distance
Commentary	This slide is about Directed Light. Read it as an occlusion decision. Decide whether the method reasons about objects, image regions, rays, or per-fragment depth comparisons. The visible cue is: - Infinity-far away imaginary light source / - Light rays run parallel and evenly throughout the scene, / direction is specified as a vector / - Light intensity does not diminish / with increasing distance
Why it matters	Visibility decides which generated candidates are actually seen from the current viewpoint.
Check yourself	Can you decide whether 'Directed Light' works per object, per image region, per ray, or per fragment?
Page 22	Local vs. Global Illumination
Source cue	- Simple case: Local lighting / - Consideration of direct illumination by light sources / (without obstacles or reflection) / - Simplified light models approximate / real lighting effect

Commentary	This slide is about Local vs. Global Illumination. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Simple case: Local lighting / - Consideration of direct illumination by light sources / (without obstacles or reflection) / - Simplified light models approximate / real lighting effect
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Local vs. Global Illumination'?

Page 23	Ambient Light
Source cue	- Undirected light rays, evenly distributed / - Light intensity constant, regardless of distance / - Simulates general room brightness (basic brightness)
Commentary	This slide is about Ambient Light. Read it as an occlusion decision. Decide whether the method reasons about objects, image regions, rays, or per-fragment depth comparisons. The visible cue is: - Undirected light rays, evenly distributed / - Light intensity constant, regardless of distance / - Simulates general room brightness (basic brightness)
Why it matters	Visibility decides which generated candidates are actually seen from the current viewpoint.
Check yourself	Can you decide whether 'Ambient Light' works per object, per image region, per ray, or per fragment?

Page 24	8.3 Material Models
Source cue	How to represent materials in computer graphics
Commentary	This slide is about 8.3 Material Models. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: How to represent materials in computer graphics
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '8.3 Material Models'?

Page 25	Material Models
Source cue	- Material models capture the different properties of materials / - Materials can have a wide variety of appearances / - Different groups of materials can be classified / based on appearance / - Metal – shiny and specular appearance
Commentary	This slide is about Material Models. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Material models capture the different properties of materials / - Materials can have a wide variety of appearances / - Different groups of materials can be classified / based on appearance / - Metal – shiny and specular appearance

Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Material Models'?
Page 26	Materials vs. Textures
Source cue	- Material models specify the material properties for an infinitesimal surface / - No material variance with respect to location is supported / (i.e., structures such as wood grain cannot be represented) / - Textures are able to support material variance with respect to location
Commentary	This slide is about Materials vs. Textures. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Material models specify the material properties for an infinitesimal surface / - No material variance with respect to location is supported / (i.e., structures such as wood grain cannot be represented) / - Textures are able to support material variance with respect to location
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Materials vs. Textures'?
Page 27	Light Material Interaction 1/3
Source cue	- Material appearance is influenced by three wavelength-dependent effects / - Reflection – light energy bounces off / - Absorption – light energy is reduced / - Transmission – light passes through the object (refraction occurs) / R = reflected energy
Commentary	This slide is about Light Material Interaction 1/3. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Material appearance is influenced by three wavelength-dependent effects / - Reflection – light energy bounces off / - Absorption – light energy is reduced / - Transmission – light passes through the object (refraction occurs) / R = reflected energy
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Material Interaction 1/3'?
Page 28	Light Material Interaction 2/3
Source cue	- Material properties determine weight of the individual effects / opaque colored material opaque black material / $A > 0$, $T = 0$, $R > 0$ $A = 1$, $T = 0$, $R = 0$ / perfect mirror material transparent material / $A = 0$, $T = 0$, $R = 1$ $A \approx 0$, $R \approx 0$, $T \approx 1$

Commentary	This slide is about Light Material Interaction 2/3. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Material properties determine weight of the individual effects / opaque colored material opaque black material / $A > 0$, $T = 0$, $R > 0$ $A = 1$, $T = 0$, $R = 0$ / perfect mirror material transparent material / $A = 0$, $T = 0$, $R = 1$ $A \approx 0$, $R \approx 0$, $T \approx 1$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Material Interaction 2/3'?
Page 29	Light Material Interaction 3/3
Source cue	- Light-material interaction described by the Spectral Response Function (SRF) / 400 Wavelength λ (nm) 700 / ygrenE / 100% / All pass Filter (ideal, real)
Commentary	This slide is about Light Material Interaction 3/3. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Light-material interaction described by the Spectral Response Function (SRF) / 400 Wavelength λ (nm) 700 / ygrenE / 100% / All pass Filter (ideal, real)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Material Interaction 3/3'?
Page 30	Reflection = SDF x SRF
Source cue	- Light-material interaction results are the product of the SDF and SRF / $100\% \times 100\%$ / SDF SRF / 0% 0% / 400 Wavelength λ (nm) 700 400 Wavelength λ (nm) 700
Commentary	This slide is about Reflection = SDF x SRF. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Light-material interaction results are the product of the SDF and SRF / $100\% \times 100\%$ / SDF SRF / 0% 0% / 400 Wavelength λ (nm) 700 400 Wavelength λ (nm) 700
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Reflection = SDF x SRF'?
Page 31	Reflection Modeling 1/2
Source cue	- All light sources can be described by their Spectral Density Functions / - Natural light: sun, fire, ... / - Artificial light: light bulb, laser, neon tube, LED, ... / - All material properties can be described by Spectral Response Functions / - Glass, Water: SRF for transmission

Commentary	This slide is about Reflection Modeling 1/2. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - All light sources can be described by their Spectral Density Functions / - Natural light: sun, fire, ... / - Artificial light: light bulb, laser, neon tube, LED, ... / - All material properties can be described by Spectral Response Functions / - Glass, Water: SRF for transmission
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Reflection Modeling 1/2'?
Page 32	Reflection Modeling 2/2
Source cue	- Synthetic and natural objects are in general not self-emitting / - Light is reflected at its surfaces, i.e., re-emitted into the scene / - Reflection is dependent on surface properties / - Rough surface: scattering of light / - Smooth surface: reflection of light
Commentary	This slide is about Reflection Modeling 2/2. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Synthetic and natural objects are in general not self-emitting / - Light is reflected at its surfaces, i.e., re-emitted into the scene / - Reflection is dependent on surface properties / - Rough surface: scattering of light / - Smooth surface: reflection of light
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Reflection Modeling 2/2'?
Page 33	8.4 Phong Illumination Model
Source cue	Determining the color of vertices
Commentary	This slide is about 8.4 Phong Illumination Model. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: Determining the color of vertices
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '8.4 Phong Illumination Model'?
Page 34	Empirical Observations
Source cue	- Light reflections are / - Dependent on the position of the observer / „... position of the observer...“ / - Dependent on the material properties of the surface / „... the specular properties of the object.“ Photo

Commentary	This slide is about Empirical Observations. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Light reflections are / - Dependent on the position of the observer / „... position of the observer...“ / - Dependent on the material properties of the surface / „... the specular properties of the object.“ Photo
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Empirical Observations'?
Page 35	Illumination Models
Source cue	- Illumination models calculate the color of scene objects / based on the following properties / - Location – Position $\mathbf{p} = (x, y, z)$ of the selected point $\mathbf{p}$ / - Orientation – normal direction $\mathbf{n} = (n_x, n_y, n_z)$ of surface in $\mathbf{p} / x y z$
Commentary	This slide is about Illumination Models. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Illumination models calculate the color of scene objects / based on the following properties / - Location – Position $\mathbf{p} = (x, y, z)$ of the selected point $\mathbf{p}$ / - Orientation – normal direction $\mathbf{n} = (n_x, n_y, n_z)$ of surface in $\mathbf{p} / x y z$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Illumination Models'?
Page 36	yaR
Source cue	thgiL / Diffuse Intensity 1/3 / - Simulation of light reflections on diffuse surfaces / - Model assumption / - Diffuse light reflections are non-directional (scattering)
Commentary	This slide is about yaR. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: thgiL / Diffuse Intensity 1/3 / - Simulation of light reflections on diffuse surfaces / - Model assumption / - Diffuse light reflections are non-directional (scattering)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'yaR'?
Page 37	Diffuse Intensity 2/3
Source cue	- Equation / $I = I_{\text{diff}} \cdot \mathbf{n} \cdot \mathbf{n} \cdot \max(0, \dots)$ / - I_{diff} diffuse material properties (surface dependent) / - I_{diff} diffuse light intensity (light source dependent)

Commentary	This slide is about Diffuse Intensity 2/3. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Equation / $I = I_{diff} \cdot \cos \theta$ / $\max(0, \dots)$ / I_{diff} diffuse material properties (surface dependent) / I_{diff} diffuse light intensity (light source dependent)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Diffuse Intensity 2/3'?

Page 38	Diffuse Intensity 3/3
Source cue	$I = I_{diff} \cdot \cos \theta$ / $\max(0, \dots)$ / I_{diff} / https://bit.ly/2XFhbcM / Extrahierte Tabellen: / [Tabelle 1]
Commentary	This slide is about Diffuse Intensity 3/3. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: $I = I_{diff} \cdot \cos \theta$ / $\max(0, \dots)$ / I_{diff} / https://bit.ly/2XFhbcM / Extrahierte Tabellen: / [Tabelle 1]
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Diffuse Intensity 3/3' into a causal sentence instead of repeating the slide title?

Page 39	Ambient Intensity 1/2
Source cue	- Simulation of the general room brightness / - Indirect light incidence or multiple reflections / - Model assumptions / - Constant influence on all scene objects / - No influence of position or alignment
Commentary	This slide is about Ambient Intensity 1/2. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Simulation of the general room brightness / - Indirect light incidence or multiple reflections / - Model assumptions / - Constant influence on all scene objects / - No influence of position or alignment
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Ambient Intensity 1/2'?

Page 40	Ambient Intensity 2/2
Source cue	$I = I_{amb}$ / I_{amb} / https://bit.ly/2XGn61n / Extrahierte Tabellen: / [Tabelle 1]
Commentary	This slide is about Ambient Intensity 2/2. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: $I = I_{amb}$ / I_{amb} / https://bit.ly/2XGn61n / Extrahierte Tabellen: / [Tabelle 1]
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.

Check yourself	Can you turn 'Ambient Intensity 2/2' into a causal sentence instead of repeating the slide title?
Page 41	Specular Intensity 1/3
Source cue	- Simulation of light reflections on specular surfaces / - Model assumption / - Reflection of light are directed (reflection) / - Observations / - Reflections of light are observer-dependent
Commentary	This slide is about Specular Intensity 1/3. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Simulation of light reflections on specular surfaces / - Model assumption / - Reflection of light are directed (reflection) / - Observations / - Reflections of light are observer-dependent
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Specular Intensity 1/3'?
Page 42	Reflection Vector ■■■
Source cue	- Calculation of the reflection vector / - Incidence angle corresponds to exit angle / - Equation / $\mathbf{r} = 2\mathbf{n} \cdot \mathbf{l} - \mathbf{l}$
Commentary	This slide is about Reflection Vector ■■■. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Calculation of the reflection vector / - Incidence angle corresponds to exit angle / - Equation / $\mathbf{r} = 2\mathbf{n} \cdot \mathbf{l} - \mathbf{l}$
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Reflection Vector ■■■' into a causal sentence instead of repeating the slide title?
Page 43	Specular Intensity 2/3
Source cue	- Equation / $I_s = \mathbf{n} \cdot \mathbf{l} \cdot \max(0, \mathbf{n} \cdot \mathbf{v})$ / ■■■■ ■■■ / - ■ specular material properties (surface dependant) / - ■ specular light intensity (light source dependent)
Commentary	This slide is about Specular Intensity 2/3. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Equation / $I_s = \mathbf{n} \cdot \mathbf{l} \cdot \max(0, \mathbf{n} \cdot \mathbf{v})$ / ■■■■ ■■■ / - ■ specular material properties (surface dependant) / - ■ specular light intensity (light source dependent)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Specular Intensity 2/3'?
Page 44	Specular Exponent ■
Source cue	- ■ influences fall off of specular highlight ($\mathbf{n} \cdot \mathbf{l}$ [1, ■] mit ■ ≈ 100) / $I_s = 1 / (\mathbf{n} \cdot \mathbf{l})^5$ / $I_s = 1 / (\mathbf{n} \cdot \mathbf{l})^5$

Commentary	This slide is about Specular Exponent n . Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - n influences fall off of specular highlight ($\frac{1}{n^2}$ [1, 2] mit $n \approx 100$) / $\frac{1}{n^2} = 1 / (n^2 \cdot x(0, \frac{1}{n^2} \cdot \frac{1}{n^2}))1 / n^2 = 5 / (n^2 \cdot x(0, \frac{1}{n^2} \cdot \frac{1}{n^2}))5$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Specular Exponent n '?

Page 45	Specular Intensity 3/3
Source cue	$I_{\text{spec}} = 2 \cdot \frac{I_{\text{diff}} \cdot I_{\text{ref}}}{I_{\text{diff}} + I_{\text{ref}}} - \frac{I_{\text{diff}} \cdot I_{\text{ref}}}{I_{\text{diff}} + I_{\text{ref}}} \cdot \frac{I_{\text{diff}}}{I_{\text{diff}} + I_{\text{ref}}}$ https://bit.ly/2XbPc7s $I_{\text{spec}} = \frac{I_{\text{diff}} \cdot I_{\text{ref}}}{I_{\text{diff}} + I_{\text{ref}}} \cdot \frac{I_{\text{diff}}}{I_{\text{diff}} + I_{\text{ref}}}$ Extrahierte Tabellen: / [Tabelle 1]
Commentary	This slide is about Specular Intensity 3/3. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: $I_{\text{spec}} = 2 \cdot \frac{I_{\text{diff}} \cdot I_{\text{ref}}}{I_{\text{diff}} + I_{\text{ref}}} - \frac{I_{\text{diff}} \cdot I_{\text{ref}}}{I_{\text{diff}} + I_{\text{ref}}} \cdot \frac{I_{\text{diff}}}{I_{\text{diff}} + I_{\text{ref}}}$ https://bit.ly/2XbPc7s $I_{\text{spec}} = \frac{I_{\text{diff}} \cdot I_{\text{ref}}}{I_{\text{diff}} + I_{\text{ref}}} \cdot \frac{I_{\text{diff}}}{I_{\text{diff}} + I_{\text{ref}}}$ Extrahierte Tabellen: / [Tabelle 1]
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Specular Intensity 3/3' into a causal sentence instead of repeating the slide title?

Page 46	Blinn-Phong Illumination Model 1/2
Source cue	- Alternative simulation of specular reflections / - Replace mirror direction $\mathbf{m}$ with Halfway-Vector $\mathbf{h}$ / - $\mathbf{h} = \frac{\mathbf{n} + \mathbf{v}}{ \mathbf{n} + \mathbf{v} }$ / - $\mathbf{h}$ is halfway between $\mathbf{n}$ and $\mathbf{v}$ / - Equation
Commentary	This slide is about Blinn-Phong Illumination Model 1/2. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Alternative simulation of specular reflections / - Replace mirror direction $\mathbf{m}$ with Halfway-Vector $\mathbf{h}$ / - $\mathbf{h} = \frac{\mathbf{n} + \mathbf{v}}{ \mathbf{n} + \mathbf{v} }$ / - $\mathbf{h}$ is halfway between $\mathbf{n}$ and $\mathbf{v}$ / - Equation
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Blinn-Phong Illumination Model 1/2'?

Page 47	Blinn-Phong Illumination Model 2/2
Source cue	$\mathbf{R} = \mathbf{R}_d + \mathbf{R}_s / \text{https://bit.ly/2ICDwBD}$ $\mathbf{R} = \mathbf{R}_d \cdot \mathbf{R}_s \cdot \mathbf{R}_s \times (0, \mathbf{R}_d \cdot \mathbf{R}_s) /$ $\mathbf{R}_d \cdot \mathbf{R}_s \cdot \mathbf{R}_s / \text{Extrahierte Tabellen: / [Tabelle 1]}$
Commentary	This slide is about Blinn-Phong Illumination Model 2/2. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: $\mathbf{R} = \mathbf{R}_d + \mathbf{R}_s / \text{https://bit.ly/2ICDwBD}$ $\mathbf{R} = \mathbf{R}_d \cdot \mathbf{R}_s \cdot \mathbf{R}_s \times (0, \mathbf{R}_d \cdot \mathbf{R}_s) /$ $\mathbf{R}_d \cdot \mathbf{R}_s \cdot \mathbf{R}_s / \text{Extrahierte Tabellen: / [Tabelle 1]}$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.

Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Blinn-Phong Illumination Model 2/2'?
Page 48	Phong Illumination Model
Source cue	- Additive combination of all factors / $I = I_a + I_r + I_r \cdot I_r$ / $I = I_a + I_r \cdot \max(0, \cos^2 \theta)$
Commentary	This slide is about Phong Illumination Model. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Additive combination of all factors / $I = I_a + I_r + I_r \cdot I_r$ / $I = I_a + I_r \cdot \max(0, \cos^2 \theta)$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Phong Illumination Model'?
Page 49	Phong Illumination Model
Source cue	- Additive combination of all factors / $I = I_a + I_r + I_r \cdot I_r$ / $I = I_a + I_r \cdot \max(0, \cos^2 \theta)$
Commentary	This slide is about Phong Illumination Model. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Additive combination of all factors / $I = I_a + I_r + I_r \cdot I_r$ / $I = I_a + I_r \cdot \max(0, \cos^2 \theta)$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Phong Illumination Model'?
Page 50	8.5 Shading
Source cue	Coloring triangles when using vertex-based illumination
Commentary	This slide is about 8.5 Shading. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: Coloring triangles when using vertex-based illumination
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by '8.5 Shading'?
Page 51	Shading
Source cue	- Illumination model usually applied in the fragment stage / - Vertex-based illumination also possible to save compute / - Requires to spread the illumination information of the vertices / over the triangle / Flat Gouraud Phong

Commentary	This slide is about Shading. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Illumination model usually applied in the fragment stage / - Vertex-based illumination also possible to save compute / - Requires to spread the illumination information of the vertices / over the triangle / Flat Gouraud Phong
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Shading'?

Page 52	Shading Computation
Source cue	- Tasks of shading procedures / - Colorize surfaces based on colors calculated by illumination model / - Shading is closely linked to the rasterization process and integrated into it / - Distribute lighting and shading / - Calculate light intensity for surface points
Commentary	This slide is about Shading Computation. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Tasks of shading procedures / - Colorize surfaces based on colors calculated by illumination model / - Shading is closely linked to the rasterization process and integrated into it / - Distribute lighting and shading / - Calculate light intensity for surface points
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Shading Computation'?

Page 53	Flat Shading
Source cue	- Operation / - Calling the lighting model for a single point of a surface / (e.g., polygon center) / - Shading of the entire polygon in the determined intensity / - Useful if the following assumptions apply
Commentary	This slide is about Flat Shading. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Operation / - Calling the lighting model for a single point of a surface / (e.g., polygon center) / - Shading of the entire polygon in the determined intensity / - Useful if the following assumptions apply
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Flat Shading'?

Page 54	Gouraud Shading 1/5
Source cue	- Shading method for polygons based on the interpolation / of intensity values at the polygon vertices / - Requirements / - Area decomposition in polygons (usually triangles) / - Normals for polygon corners (vertex normals)

Commentary	This slide is about Gouraud Shading 1/5. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Shading method for polygons based on the interpolation / of intensity values at the polygon vertices / - Requirements / - Area decomposition in polygons (usually triangles) / - Normals for polygon corners (vertex normals)
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Gouraud Shading 1/5'?

Page 55	Gouraud Shading 2/5
Source cue	- Proceeding / - Compute intensity values ■ for polygon corners ■ / ■ ■ / - Linear interpolation of the intensity values along the edges / - Linear interpolation of the intensity values along the scanlines
Commentary	This slide is about Gouraud Shading 2/5. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Proceeding / - Compute intensity values ■ for polygon corners ■ / ■ ■ / - Linear interpolation of the intensity values along the edges / - Linear interpolation of the intensity values along the scanlines
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Gouraud Shading 2/5'?

Page 56	Gouraud Shading 3/5
Source cue	- Shading at conceptually smooth transitions / - Geometrically "hard" edges instead of conceptually "soft" edges / - Gouraud shading can create visually smooth transitions / through smooth shading / - Example: ■ / ■ smooth transition, ■ / ■ hard transition
Commentary	This slide is about Gouraud Shading 3/5. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Shading at conceptually smooth transitions / - Geometrically "hard" edges instead of conceptually "soft" edges / - Gouraud shading can create visually smooth transitions / through smooth shading / - Example: ■ / ■ smooth transition, ■ / ■ hard transition
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Gouraud Shading 3/5'?

Page 57	Gouraud Shading 4/5
Source cue	- interpolation problems / - The shading inside a triangle results from interpolation / - Intensity fluctuations in the interior are not taken into account / - Intensity fluctuations can only be considered by finer tessellation / - Example: Spotlight shines into the interior of a triangle

Commentary	This slide is about Gouraud Shading 4/5. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - interpolation problems / - The shading inside a triangle results from interpolation / - Intensity fluctuations in the interior are not taken into account / - Intensity fluctuations can only be considered by finer tessellation / - Example: Spotlight shines into the interior of a triangle
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Gouraud Shading 4/5'?

Page 58	Gouraud Shading 5/5
Source cue	- Interpolation problems / - Contiguous polygons that do not fully share the same edges / will be shaded differently
Commentary	This slide is about Gouraud Shading 5/5. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Interpolation problems / - Contiguous polygons that do not fully share the same edges / will be shaded differently
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Gouraud Shading 5/5'?

Page 59	Phong Shading 1/2
Source cue	- Interpolation of the corner normals instead of the corner intensities / - Proceeding / - Calculation of the corner normals / - Interpolation of the normals between the corners / - Interpolation between the endpoints of a scanline
Commentary	This slide is about Phong Shading 1/2. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Interpolation of the corner normals instead of the corner intensities / - Proceeding / - Calculation of the corner normals / - Interpolation of the normals between the corners / - Interpolation between the endpoints of a scanline
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Phong Shading 1/2'?

Page 60	Phong Shading 2/2
Source cue	- Advantages / - Pixel-based (i.e., image-precise) evaluation of the light sources / - Highly focused highlights at corners are not interpolated over the edge / - Highlights inside a triangle can be displayed / - Disadvantage

Commentary	This slide is about Phong Shading 2/2. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Advantages / - Pixel-based (i.e., image-precise) evaluation of the light sources / - Highly focused highlights at corners are not interpolated over the edge / - Highlights inside a triangle can be displayed / - Disadvantage
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Phong Shading 2/2'?

Page 61	Literature and other sources used in this chapter
Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Commentary	This slide is about Literature and other sources used in this chapter. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Literature and other sources used in this chapter' into a causal sentence instead of repeating the slide title?

Page 62	Lesen und Ausprobieren
Source cue	- Zum Lesen - Zum Ausprobieren / Marschner, Steve und Peter Shirley: / - Simulation von drei / Fundamentals of Computer Graphics, / ausgewählten Materialien
Commentary	This slide is about Lesen und Ausprobieren. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Zum Lesen - Zum Ausprobieren / Marschner, Steve und Peter Shirley: / - Simulation von drei / Fundamentals of Computer Graphics, / ausgewählten Materialien
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Lesen und Ausprobieren'?

08.1 Physics of Light

Source location: Section 8.1

Commentary

The physics slides provide intuition for light as energy interacting with surfaces. For this course, the goal is not full physical simulation but a usable model for real-time rendering.

Important terms are reflection, absorption, wavelength/color, intensity, and direction. These terms become shader inputs or material parameters later. The simplified model must still preserve the relation between light direction, surface orientation, and observed brightness.

Mental model

Shading is an approximation of light-surface interaction that is cheap enough for rendering.

Check yourself

Why does surface orientation affect diffuse brightness?

08.2 Light Sources

Source location: Section 8.2

Commentary

Light source models define where light comes from and how its direction and intensity are computed. Directional lights approximate distant sources with parallel rays. Point lights emit from a position and often include attenuation. Spotlights add a directional cone.

In shader code, the type of light determines how you compute the light vector. A directional light can use a fixed direction; a point light requires subtracting the surface position from the light position.

Mental model

Different light types mainly change how the light direction and intensity are computed at the surface point.

Check yourself

What is the difference between a directional light vector and a point light vector?

08.3 Material Models

Source location: Section 8.3

Commentary

A material model tells the shader how a surface responds to light. Diffuse material scatters light broadly; specular material creates view-dependent highlights; ambient terms approximate background illumination.

Material coefficients are not arbitrary decoration. They scale the contribution of each lighting term. If the specular coefficient is zero, the surface should not show a specular highlight under that model.

Mental model

Light asks what arrives; material answers how the surface reacts.

Check yourself

Which material parameter would you adjust to reduce shiny highlights?

08.4 Phong Illumination Model

Source location: Section 8.4

Commentary

The Phong illumination model combines ambient, diffuse, and specular terms. Diffuse lighting uses the angle between normal and light direction. Specular lighting also depends on the viewer because highlights move with the view direction.

The formula is less important than the decomposition. Ambient is a rough constant approximation. Diffuse is orientation-dependent matte reflection. Specular is view-dependent shininess. A correct answer should name the vectors and normalize them.

Mental model

Phong-style lighting is a sum of simple terms, each modeling a different visual effect.

Check yourself

Why does the specular term depend on the view direction?

08.5 Shading

Source location: Section 8.5

Commentary

Shading decides where the illumination calculation is evaluated and how results are interpolated. Flat shading uses one value per primitive. Gouraud shading evaluates at vertices and interpolates colors. Phong shading interpolates normals and evaluates lighting per fragment.

Per-fragment shading is more expensive but can represent highlights more accurately. This connects directly to the pipeline: interpolation during rasterization provides the data used by the fragment shader.

Mental model

The shading method decides what is computed at vertices and what is computed at fragments.

Check yourself

Why can Gouraud shading miss a small specular highlight?

End-of-lecture summary

If you remember only one thing from Lecture 08, remember this: This lecture explains how visible surface points receive color from local light, material, normal, and view information. It is about shading a point, not solving all global light transport.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.